Servicenow

Session Agenda

- Session Agenda: Employee Asset Management System
- Introduction to Tables
- Table Extension
- Extensible Tables
- Auto Number Control
- Fields (Columns)
- Reference Field
- Dictionary
- Dictionary Override
- Validation
- Form Design & Form Builder
- Views
- List Layout
- Filters
- Remote Table
- Schema Map
- Modules & Application Menu
- End-to-End Example

Table - Explanation

- A table is the core data structure in ServiceNow.
- It stores records (rows) and fields (columns).
- Every application depends on tables.

Table - Requirement

- Create Asset Request table to store asset data.

Table - Deep Details

- Steps:
- System Definition → Tables → New
- Label: Asset Request
- Name: u_asset_request
- Extends: Task

- Why extend Task?
- Reuse built-in fields like number, state.

- Other methods:
- - Import Excel
- - XML import

Fields (Columns) - Explanation

- Fields define what data is stored inside the table.
- Each field has a type and properties.

Fields (Columns) - Requirement

- Create fields:
- Asset Name, Asset Type, Assigned To, Age

Fields (Columns) - Deep Details

- Methods:
- 1. Table Columns
- 2. Form Design
- 3. Form Builder

- Types:
- - String
- - Integer
- - Choice
- - Reference

- Example:
- Assigned To → Reference → sys_user

Reference Field - Explanation

- Reference field links one table to another.
- It creates relationships between data.

Reference Field - Requirement

- Assign asset to employee using reference field.

Reference Field - Deep Details

- Steps:
- Create field → Type: Reference
- Select table: sys_user

- Result:
- Dropdown list of users

- Use cases:
- - Assignment
- - Ownership
- - Approvals

Dictionary - Explanation

- Dictionary controls all field behavior.
- It is the backend configuration of fields.

Dictionary - Requirement

- Make Asset Name mandatory and set default values.

Dictionary - Deep Details

- Steps:
- Right click field → Configure Dictionary

- Options:
- - Mandatory
- - Default value
- - Max length
- - Read only

- Advanced:
- Validation rules

Dictionary Override - Explanation

- Used to modify parent table fields in child table.

Dictionary Override - Requirement

- Change Task field behavior only in Asset table.

Dictionary Override - Deep Details

- Steps:
- Open Dictionary → Override

- Example:
- Make 'State' read-only in Asset table only

- Use:
- Control inherited fields

Validation - Explanation

- Validation ensures correct data input.

Validation - Requirement

- Restrict Age from 1 to 9.

Validation - Deep Details

- Methods:
- 1. Integer → Min=1 Max=9
- 2. Regex → ^[1-9]$
- 3. Client Script (advanced)

- Why?
- Prevent invalid data

Form Design - Explanation

- Form is used to insert and update data.
- Form Design controls layout.

Form Design - Requirement

- Design sections for Asset form.

Form Design - Deep Details

- Steps:
- Configure → Form Design

- Sections:
- - Asset Info
- - Assignment
- - Approval

- Layout:
- 1 or 2 columns

- Alternative:
- Form Builder (modern UI)

Views - Explanation

- Views provide different UI for different users.

Views - Requirement

- Create Manager View for approvals only.

Views - Deep Details

- Steps:
- Form Design → New View

- Example:
- Manager sees only approval fields

- Use:
- Role-based UI

List Layout - Explanation

- List view shows table records.
- Layout controls visible columns.

List Layout - Requirement

- Show only important columns in list.

List Layout - Deep Details

- Steps:
- Right click → Configure → List Layout

- Add/remove columns
- Reorder columns

- Use:
- Improve usability

Filters - Explanation

- Filters display specific records.

Filters - Requirement

- Show only approved assets.

Filters - Deep Details

- Steps:
- Filter → Status = Approved

- Advanced:
- Multiple conditions

- Use:
- Reporting
- Data analysis

Auto Number - Explanation

- Automatically generates unique record IDs.

Auto Number - Requirement

- Create numbering like AST0001.

Auto Number - Deep Details

- Steps:
- Number Maintenance

- Configure:
- Prefix
- Starting number

- Use:
- Unique identification

Extensible Table - Explanation

- Controls if table can be extended.

Extensible Table - Requirement

- Prevent other tables from inheriting.

Extensible Table - Deep Details

- Set Extensible = false

- Use:
- Protect system design

Remote Table - Explanation

- Temporary table for integration data.

Remote Table - Requirement

- Receive data from external system.

Remote Table - Deep Details

- Flow:
- External → Remote → Validate → Main

- Use:
- Data safety
- Integration control

Schema Map - Explanation

- Visual representation of table relationships.

Schema Map - Requirement

- View parent-child relationship.

Schema Map - Deep Details

- Open Schema Map

- Shows:
- - Parent table
- - Child table

- Use:
- Understand structure

Modules & Application - Explanation

- Defines navigation structure in ServiceNow.

Modules & Application - Requirement

- Create Asset Management app.

Modules & Application - Deep Details

- Steps:
- Create Application Menu
- Add Module

- Example:
- App: Asset Management
- Module: Asset Requests

Practice

Task 1

Prevent Table Extension

Create table: Restaurant Order

Extends: Task

Set Extensible = false

Go to: System Definition → Tables → New

Set name: u_restaurant_order

Check 'Extends' and choose: task

Uncheck the 'Extensible' checkbox

Save — no other table can extend this table

Task 2

Create Child Table

Create a child table that extends Restaurant Order

Table name: u_delivery_order

Label: Delivery Order

Extends: u_restaurant_order (Restaurant Order)

Hierarchy:

task → u_restaurant_order → u_delivery_order

Go to: System Definition → Tables → New

Set Extends = u_restaurant_order

Task 3

Create Fields

Add the following fields to Restaurant Order:

- Dish Name → Type: String
- Quantity → Type: Integer
- Table Number → Type: Integer

Method 1 — Table Columns:

System Definition → Tables → Restaurant Order → Columns tab → New

Method 2 — Form Design:

Right-click form header → Configure → Form Design → Drag field

Task 4

Reference Field

Add a reference field to link the order to a waiter

- Field Label : Assigned Waiter
- Column Name : u_assigned_waiter
- Type : Reference
- References : sys_user
- Display Field: name

Go to: Table Columns → New → Type = Reference → Table = sys_user

Task 5

Dictionary Configuration

Control field behavior via the Dictionary

- Dish Name → Mandatory = true, Max Length = 100
- Quantity → Default Value = 1
- Table Number→ Max Length = 3

Navigate to: System Definition → Dictionary

Filter by table: u_restaurant_order

Open each field and set the required attributes

Task 6

Dictionary Override

Modify the inherited 'State' field only for Restaurant Order

- Field : State (inherited from task)
- Changes: Read-only = true, Mandatory = true
- Scope : Only affects u_restaurant_order

Steps:

System Definition → Dictionary → filter by u_restaurant_order

Find 'state' field → click Override

Set Read-only and Mandatory → Save

Task 7

Validation

Restrict Quantity field to values between 1 and 9

Method 1 — Integer Min/Max:

Open Dictionary entry for Quantity

Set Minimum Value: 1

Set Maximum Value: 9

Method 2 — Regex:

Open Dictionary → Validation tab

Regex pattern: ^[1-9]$

Error message: "Quantity must be between 1 and 9"

Task 8

Form Design

Design the form with 3 sections and 2-column layout

Section 1 — Order Info:

Dish Name, Quantity, Table Number, Assigned Waiter

Section 2 — Customer Info:

Customer Name, Phone Number

Section 3 — Payment:

Payment Method, Total Amount

Right-click form → Configure → Form Design → Add Sections

Task 9

Views

Create role-based form views

Chef View:

Fields: Dish Name, Quantity

Manager View:

Fields: Payment, Status

Steps:

Right-click form → Configure → Form Layout

Click 'New View' → name it (e.g. Chef)

Add the relevant fields → Save

Task 10

List Layout

Customize the list view to show specific columns

Columns to display:

- Dish Name (u_dish_name)
- Quantity (u_quantity)
- Status (state)

Steps:

Open Restaurant Order list

Right-click any column header

Configure → List Layout

Move fields to Selected using the slushbucket → Save

Task 11

Filters

Filter the list to show specific records

Filter 1: Status = Pending

Field: State | Operator: is | Value: Pending (1)

Filter 2: Table Number = 5

Field: u_table_number | Operator: is | Value: 5

Combined URL query:

state=1^u_table_number=5

Save filter as a favorite for quick access

Task 12

Auto Number

Generate order IDs automatically in format ORD0001

Steps:

Go to: System Definition → Number Maintenance → New

Table: u_restaurant_order

Prefix: ORD

Starting Number: 0001

Increment: 1

Save

Result: ORD0001, ORD0002, ORD0003 ...

Task 13

Remote Table

Handle external orders via Remote Table integration

Use case: Simulate a delivery app sending orders to ServiceNow

Steps:

- System Definition → Tables → New
- Check 'Remote' checkbox
- Define remote table name and field mapping
- Set REST Message or MID Server endpoint
- Use Transform Map to map external fields to local fields

Data stays in external system — ServiceNow reads/writes via API

Task 14

Schema Map

Visualize table relationships in the Schema Map

Tables shown:

- task (base system table)
- u_restaurant_order (extends task)
- u_delivery_order (extends u_restaurant_order)

Also shows:

- Reference: u_assigned_waiter → sys_user

Navigation:

System Definition → Tables & Columns → Schema Map

Select: u_restaurant_order → View relationships

Task 15

Application & Module

Create an Application and a Module for navigation

Application:

Name : Restaurant Management

Scope: x_[prefix]_restaurant

Go to: System Applications → Applications → New

Module:

Application: Restaurant Management

Title : Orders

Link Type : List of Records

Table : u_restaurant_order

Go to: System UI → Application Menus → Modules → New

Task 16 🚀

Final Challenge — Full System Build

1. Create Table → u_restaurant_order
2. Extend from Task → Extensible = false
3. Create Child Table → u_delivery_order
4. Add Fields → Dish Name, Quantity, Table Number
5. Add Reference Field → Assigned Waiter → sys_user
6. Configure Dictionary → Mandatory, Default, Max Length
7. Apply Override → State: Read-only + Mandatory
8. Add Validation → Quantity range 1–9
9. Design Form → 3 Sections, 2 Columns
10. Create Views → Chef View + Manager View
11. Configure List Layout → Dish Name, Quantity, Status
12. Add Filters → Status=Pending, Table Number=5
13. Configure Auto Number → Format: ORD0001

Homework

Table
Department

Delete | Update | Delete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info

* Label: Department
* Name: u_department
Application: Global
Remote Table:

Table
Department

Delete | Update | Delete All Records

Columns | Controls | Application Access

Table Columns | Column label | Search | 1 to 11 of 11 | New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false
Department Code	String	(empty)	100		false
Department Name	String	(empty)	100		false
Floor Number	Integer	(empty)	40		false
HOD Name	String	(empty)	100		false
Number	String	(empty)	40	javascript:global.getNextObjNumberPadded();	false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updated	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false

Department Table

1. Department Name
2. Department Code
3. Floor Number
4. HOD Name

Table
Doctor

Delete | Update | Delete All Records

processes. More Info

* Label: Doctor
* Name: u_doctor

Application: Global

Remote Table

Table
Doctor

Column label	Type	Reference	Max length
Availability	Choice	(empty)	40
Created	Date/Time	(empty)	40
Created by	String	(empty)	40
Department	String	Department	40
Doctor ID	String	(empty)	100
Doctor Name	String	(empty)	100
Email	String	(empty)	100
Number	String	(empty)	40
Phone	String	(empty)	100
Specialization	String	(empty)	200
Sys ID	Sys ID (GUID)	(empty)	32
Updated	Date/Time	(empty)	40
Updated by	String	(empty)	40
Updates	Integer	(empty)	40

Insert a new row...

Doctor Table

1. Doctor Name
2. Doctor ID
3. Specialization
4. Phone
5. Email
6. Department (Reference)
7. Availability

Table
Patient

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record processes. More Info

* Label: Patient
* Name: u_patient

Table
Patient

×	Address	String	(empty)	1,000
×	Admitted Date	Date	(empty)	40
×	Age	Integer	(empty)	40
×	Blood Group	Choice	(empty)	40
	Created	Date/Time	(empty)	40
	Created by	String	(empty)	40
×	Gender	Choice	(empty)	40
×	Number	String	(empty)	40
×	Patient ID	String	(empty)	100
×	Patient Name	String	(empty)	100
×	Phone	String	(empty)	100
	Sys ID	Sys ID (GUID)	(empty)	32
	Updated	Date/Time	(empty)	40
	Updated by	String	(empty)	40
	Updates	Integer	(empty)	40

Patient Table

1 Patient Name
2 Patient ID
3 Age
4 Gender
5 Phone
6 Address
7 Blood Group
8 Admitted Date

Table
Room

A table is a collection of records in the database. Each record corresponds to a row in a table, and processes. More Info

* Label: Room
* Name: u_room

Table
Room

Dictionary Entries

	Column label	Type	Reference	Max length
×	Availability	Choice	(empty)	40
×	Charges Per Day	Decimal	(empty)	20
	Created	Date/Time	(empty)	40
	Created by	String	(empty)	40
×	Floor	Integer	(empty)	40
×	Number	String	(empty)	40
×	Room Number	String	(empty)	100
×	Room Type	Choice	(empty)	40
	Sys ID	Sys ID (GUID)	(empty)	32
	Updated	Date/Time	(empty)	40
	Updated by	String	(empty)	40
	Updates	Integer	(empty)	40

Room Table

1 Room Number
2 Room Type
3 Floor
4 Availability
5 Charges Per Day

Table
Appointment

Delete | Update | Delete All Records

table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info

* Label: Appointment
* Name: u_appointment

Application: Global

Remote Table

Table
Appointment

Dictionary Entries

Column label	Type	Reference	Max length
Appointment Number	String	(empty)	100
Created	Date/Time	(empty)	40
Created by	String	(empty)	40
Date	Date	(empty)	40
Department	String	Department	40
Doctor	String	Doctor	40
Number	String	(empty)	40
Patient	String	Patient	40
Status	Choice	(empty)	40
Sys ID	Sys ID (GUID)	(empty)	32
Time	Time	(empty)	40
Updated	Date/Time	(empty)	40
Updated by	String	(empty)	40
Updates	Integer	(empty)	40

Appointment Table

1. Appointment Number
2. Patient (Reference)
3. Doctor (Reference)
4. Department (Reference)
5. Date
6. Time
7. Status

servicenow All Favorites History Workspaces Table - Billing Search

Table Billing

Delete

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Application processes. More Info

* Label: Billing
* Name: u_billing

Application: Global

Remote Table

servicenow All Favorites History Workspaces Table - Billing

Table Billing

Dictionary Entries

Column label	Type	Reference	Max length
Appointment	String	Appointment	40
Billing Number	String	100	40
Created	Date/Time	(empty)	40
Created by	String	(empty)	40
Number	String	(empty)	40
Patient	String	Patient	40
Payment Date	Date	(empty)	40
Payment Status	Choice	(empty)	40
Room	String	Room	40
Sys ID	Sys ID (GUID)	(empty)	32
Total	Decimal	(empty)	20
Updated	Date/Time	(empty)	40
Updated by	String	(empty)	40
Updates	Integer	(empty)	40

Billing Table

1. Bill Number
2. Patient (Reference)
3. Appointment (Reference)
4. Room (Reference)
5. Total Amount
6. Payment Status
7. Payment Date

Departments | Number | Search | Actions on selected rows... | New

All

Number	Department Code	Department Name	Floor Number	HOD Name
DEP0001002	Code 2	Dep 2	2	Test
DEP0001001	Code 1	Dep 1	1	Test

servicenow | All | Favorites | History | Application Menu - Hospital Manag... | Search

Application Menu
Hospital Managment

Update | Delete

[SN Utils] Versions (1)

Modules | for text | Search | Actions on selected rows... | New

Application menu = Hospital Managment

Title	Table	Active	Filter	Order	Link type	Device type	Roles	Updated
Patient	Patient [u_patient]	true			List of Records			2026-04-04 05:53:56
Departments	Department [u_department]	true			List of Records		u_department_user	2026-04-04 03:52:33
Billing	Billing [u_billing]	true			List of Records			2026-04-04 05:55:26
Appointment	Appointment [u_appointment]	true			List of Records			2026-04-04 05:54:58
Doctor	Doctor [u_doctor]	true			List of Records			2026-04-04 05:52:24
Room	Room [u_room]	true			List of Records			2026-04-04 05:54:23

All

Number	Availability	Department	Doctor ID	Doctor Name	Email	Phone	Specialization
DOC0001008	Available	DEP0001001	123	doctor 1	test@test.com	123456	test
DOC0001009	Available	DEP0001002	2324	Doctor 2		34567	test

Patients | Number | Search | Actions on selected rows... | New

All

Number	Address	Admitted Date	Age	Blood Group	Gender	Patient ID	Patient Name	Phone
PAT0001001	test	2026-04-04	23	A	Female	231	Patient 1	
PAT0001002	test	2026-04-09	22	B	Male	2323	patient 2	12313

Billings | Number | Search | Actions on selected rows... | New

All

Number	Appointment	Billing Number	Patient	Payment Date	Payment Status	Room	Total
BIL0001001	APP0001002	12	PAT0001002	2026-04-04	Pending	ROO0001002	1,023

Appointments | Number | Search | Actions on selected rows... | New

All

Number	Appointment Number	Date	Department	Doctor	Patient	Status	Time
APP0001002	2	2026-04-15	DEP0001002	DOC0001009	PAT0001002	Pending	10:00:00
APP0001001	1	2026-04-04	DEP0001002	DOC0001008	PAT0001001	Pending	00:00:00

servicenow All Favorites History Workspaces

Roles Name Search

All > Name >= hospital

Name	Description
Search	Search
hospital_admin	hospital_admin
hospital_billing	hospital_billing
hospital_doctor	hospital_doctor
hospital_receptionist	hospital_receptionist

servicenow All Favorites History Workspaces Groups

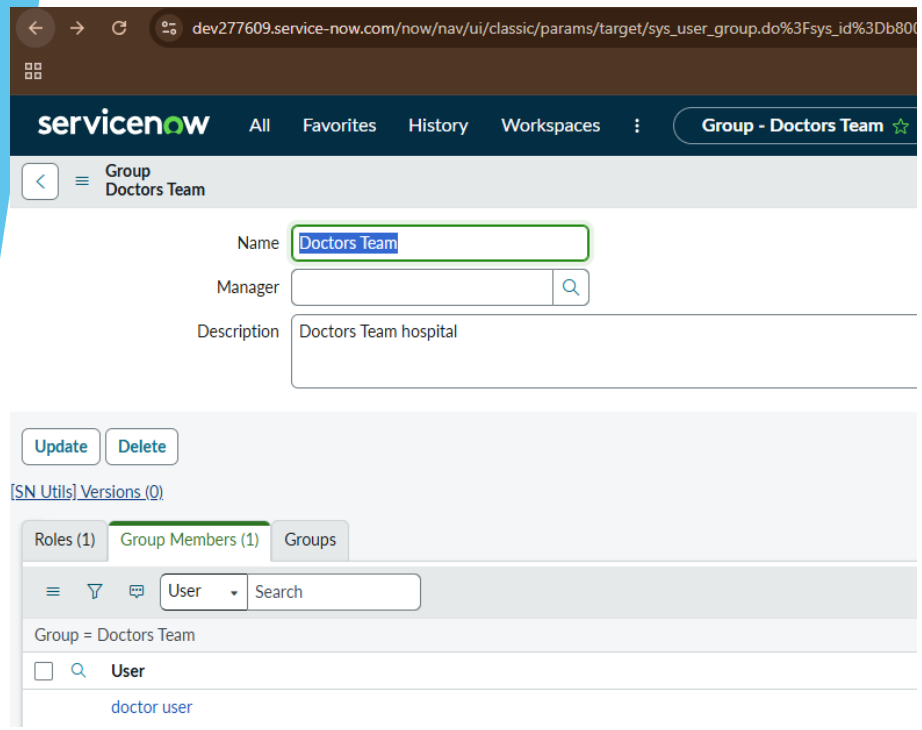
Application scope: Global
Update set: Default [Global]

Groups Description Search

Actions on selected rows... New

All > Description contains hos

Name	Description	Active	Manager	Parent	Updated
Search	*hos	Search	Search	Search	Search
Admin Team	Admin Team hospital	true	(empty)	(empty)	2026-04-04 07:21:40
Billing Team	Billing Team hospital	true	(empty)	(empty)	2026-04-04 07:24:50
Doctors Team	Doctors Team hospital	true	(empty)	(empty)	2026-04-04 07:22:38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Group - Doctors Team
Group
Doctors Team
Name
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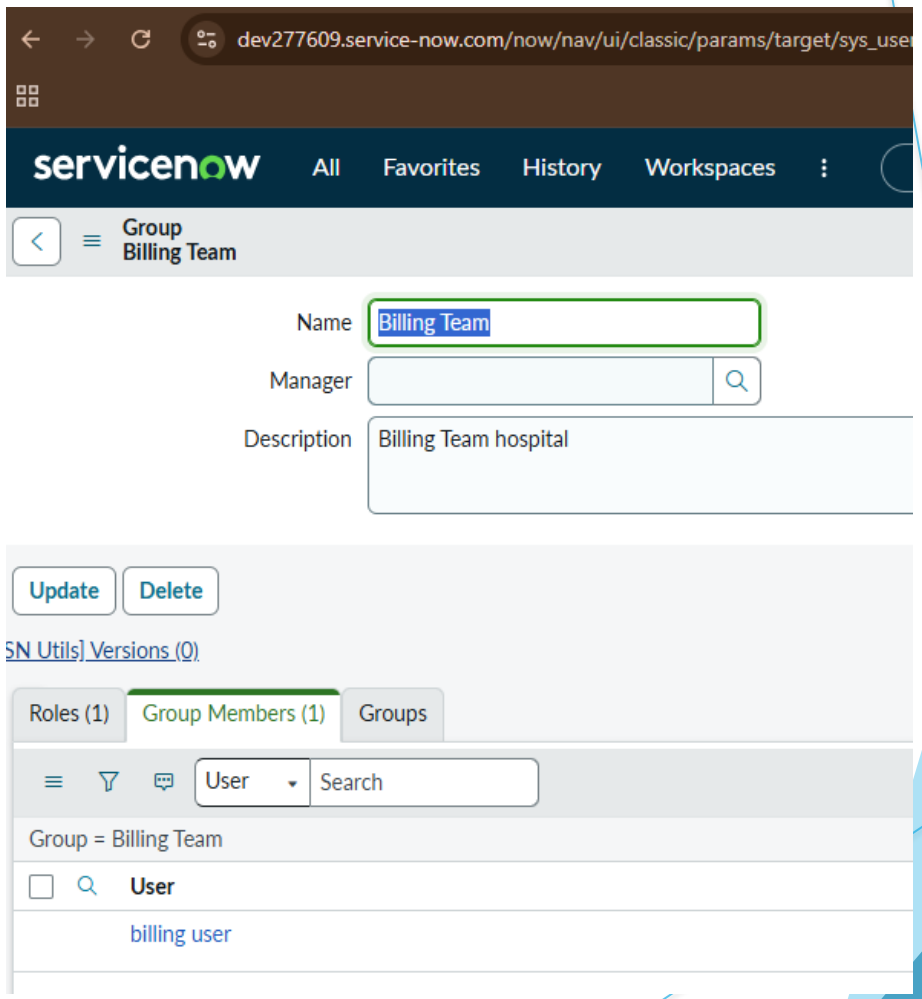
Manager
Description
Doctors Team hospital
Update
Delete
[SN Utils] Versions (0)
Roles (1)
Group Members (1)
Groups
User
Search
Group = Doctors Team
User
doctor user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Description
Billing Team hospital
Update
Delete
SN Utils] Versions (0)
Roles (1)
Group Members (1)
Groups
User
Search
Group = Billing Team
User
billing user

Questions About (Users-Groups-Roles)

What tool is used to change the layout of a form by dragging and dropping fields, sections, and annotations?

A) Form Layout
B) Form Designer
C) Dictionary Override
D) Schema Map

Answer: B

- The Form Designer provides a visual interface to drag and drop fields and sections.

Which field type allows you to create a relationship between the current table and a record in another table?

A) Choice
B) String
C) Reference
D) Journal

Answer: C

- A Reference field creates a relationship by storing a link to a record in a target table.

What is the correct syntax to open the list of all tables in the ServiceNow instance from the Filter Navigator?

A) tables.list
B) all_tables.do
C) dictionary.LIST
D) sys_db_object.LIST

Answer: D

- Typing sys_db_object.LIST in the navigator opens the table of all table definitions.

A collection of records in ServiceNow is referred to as what?

A) A Field
B) A Form
C) A Table
D) A List

Answer: C

- In ServiceNow, a table is a collection of records, while fields are the individual pieces of data within those records.

When you type <table_name>.form into the Filter Navigator, what is the result?

A) It displays a blank form to create a new record in that table
B) It opens the Dictionary for that table.
C) It displays the list of all records in that table.
D) It opens the Form Designer for that table.

Answer: A

- Using the ".form" suffix is a shortcut to open a new record entry form for that specific table.

Which of these is a unique 32-character identifier assigned to every record in ServiceNow?

A) Record ID
B) sys_id
C) Table Key
D) Index ID

Answer: B

- The sys_id is the globally unique identifier for every record in the database.

In a List view, what do the columns correspond to?

A) Records
B) Tables
C) Sections
D) Fields

Answer: D

- In a list view, rows represent records and columns represent fields.

What is the primary purpose of a Dictionary Override in ServiceNow?

A) To create a new field on a child table.

B) To allow a field on a child table to have different properties than the same field on its parent table.

C) To delete a field from a parent table without affecting child tables.

D) To move a field from a child table to a parent table

Answer: B

- Dictionary Overrides allow child tables to define unique behavior (like being mandatory or having a different default value) for fields inherited from a parent.

The Incident table extends the Task table. If you want to make the "Short description" field mandatory ONLY on the Incident table, which method is the ServiceNow best practice?

A) Modify the field directly in the Task table dictionary.

B) Create a Client Script on the Task table.

C) Create a Dictionary Override for the Incident table.

D) Create a new field on the Incident table named "Incident Short description"

Answer: C

- Making a field mandatory at the Task (Parent) level would affect all other child tables (Problem, Change, etc.). A Dictionary Override targets only the specific child table

Which of the following field attributes CANNOT be changed using a Dictionary Override?

A) Default value
B) Reference qualifier
C) Read-only status
D) Field Name

Answer: D

- You can override attributes like default values, reference qualifiers, and mandatory/read-only status, but the Field Name is part of the core database schema and remains the same across the hierarchy.

In the ServiceNow inheritance model, what happens to the fields of a parent table when it is extended by a child table?

A) The child table inherits all fields from the parent table

B) The child table gets its own independent copies of the parent's fields.

C) The child table only inherits the sys_id field.

D) Fields are moved from the parent table to the child table.

Answer: A

- Table inheritance ensures that a child table (like Incident) contains all the fields of its parent (like Task) plus any fields unique to the child.

When a table extends another table, what is the parent table often called?

A) Core Table
B) Base Table
C) Abstract Table
D) Primary Table

Answer: B

- In ServiceNow, the table being extended is referred to as the Base Table or Parent Table (e.g., Task is the base table for Incident).

Which of the following is a true statement regarding Table Inheritance?

A) A table can extend multiple parent tables simultaneously.

B) Business Rules and ACLs on a parent table do not apply to the child table.

C) A child table inherits the fields, Business Rules, and ACLs of its parent table.

D) Once a table is created, it can never be extended

Answer: C

- Inheritance includes not just fields but also platform logic like Business Rules and Security (ACLs) defined on the parent.

Thank you